

AST 6007-01 Spacecraft Dynamics and Control

Midterm Examination

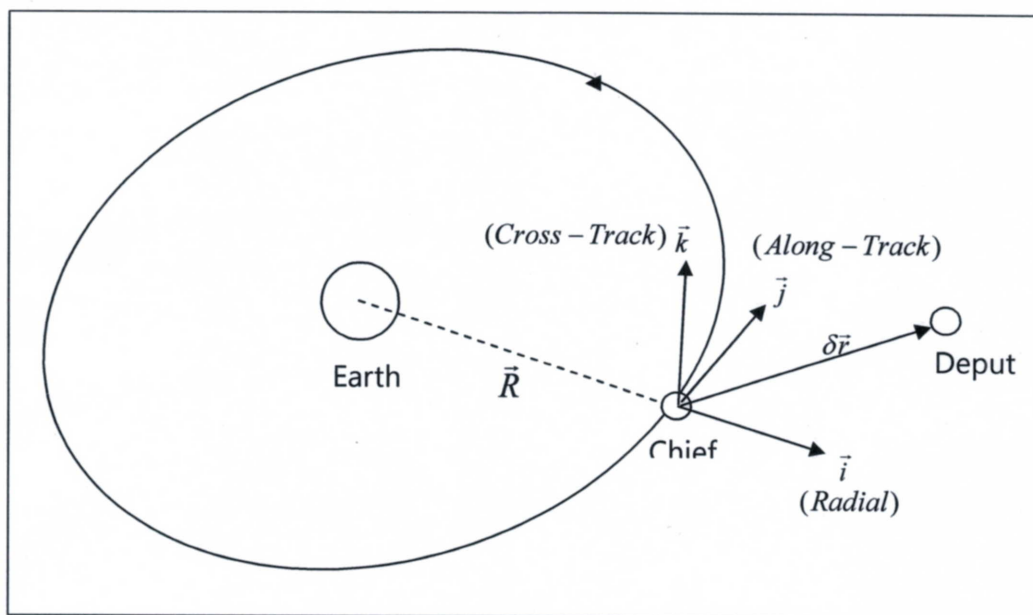
October 21, 2013 3:00-4:40 PM

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1. (30) Consider a local-vertical-local-horizontal (LVLH) $\hat{i}\hat{j}\hat{k}$ -frame attached to a circular reference orbit whose constant orbital frequency is given by $\vec{\omega} = \omega\hat{k}$.



- a. (15) Let the position of the deputy spacecraft be given by

$$\vec{r} = \vec{R} + \delta\vec{r} = (R+x)\hat{i} + y\hat{j} + z\hat{k}.$$

Use the following Newton's Law

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{u}$$

to derive the equations of motion in the $\hat{i}\hat{j}\hat{k}$ -frame. Here μ is the gravitational parameter of the Earth, $\vec{u} = [u_x \ u_y \ u_z]^T$ is the thrust acceleration vector, and $r = \sqrt{(R+x)^2 + y^2 + z^2}$ is the distance of the deputy spacecraft from the Earth center.

- b. (15) Linearize the equations of motion derived in 1-(a) with respect to a circular reference trajectory to derive the Clohessy-Wiltshire equation.

2. (30) Suppose that a spacecraft with mass $m=100\text{kg}$ operates near a circular orbit with orbital radius $r=6,578\text{km}$. The Clohessy-Wiltshire equations can describe its orbital perturbation dynamics in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. Under some conditions, the Clohessy-Wiltshire equations can be approximated by

$$\begin{cases} m\Delta\ddot{x} = F_x \\ m\Delta\ddot{y} = F_y \\ m\Delta\ddot{z} = F_z \end{cases}$$

Here $(\Delta x, \Delta y, \Delta z)$ are perturbations of the spacecraft position from the circular orbit, and (F_x, F_y, F_z) are propulsive force components expressed in the LVLH $\hat{i}\hat{j}\hat{k}$ -frame. This approximation is often acceptable over a sufficiently small time period $T > 0$.

- (15) Estimate how small the maneuver time period T should be so that the above equations are a satisfactory approximation of the Clohessy-Wiltshire equations within accuracy of about 3%. Hint: Compare the equations of motion derived in Problem 1-(b) with the given approximation.
 - (15) Using the above approximation, a two-impulse rendezvous maneuver is to transfer the initial spacecraft position vector $\Delta\vec{r}(0) = -10\hat{i}\text{km}$ and initial velocity vector $\Delta\dot{\vec{r}}(0) = 0$ to the terminal spacecraft position vector $\Delta\vec{r}(60\text{sec}) = 0$ and terminal velocity vector $\Delta\dot{\vec{r}}(60\text{sec}) = 0$. What is the change in spacecraft velocity vector that occurs at each of the impulsive burns?
3. (25) The equations of motion for the Moon-Earth circular restricted three-body problem is given as follows:

$$\begin{cases} \ddot{x} - 2\omega_0\dot{y} - \omega_0^2x = -\frac{\mu_E(x-D_E)}{r_E^3} - \frac{\mu_M(x+D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0\dot{x} - \omega_0^2y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}, \begin{cases} r_E^2 = (x-D_E)^2 + y^2 + z^2 \\ r_M^2 = (x+D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601\text{km}^3/\text{sec}^2 \\ \mu_M = 4,887\text{km}^3/\text{sec}^2 \\ D_E = 4,674\text{km} \\ D_M = 380,073\text{km} \\ \omega_0 = 2.66199 \times 10^{-6}\text{rad/sec} \end{cases}$$

- (10) Show that $(x_e = -444,645.6, y_e = 0, z_e = 0)$ is an equilibrium solution with no propulsive forces.
- (15) Linearize the equations of motion with respect to the equilibrium solution given in (a).

Remember

$$\omega_0^2 x_e = \frac{-\mu_E(x_e - D_E)}{(x_e - D_E)^3} - \frac{\mu_M(x_e + D_M)}{(x_e + D_M)^3}$$

4. (15) Discuss the following concepts briefly and precisely.

- (5) Halo Orbit (Original Definition in Celestial Mechanics)
- (10) Rendezvous/Orbit Transfer/Interception Maneuvers

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$$\Delta\ddot{x} - 2\omega_0\Delta\dot{y} - \omega_0^2(x_e) - \omega_0^2\Delta x$$

$$= \frac{-\mu_E}{(x_e - D_E)^2} - \frac{\mu_M}{(x_e + D_M)^2} + \frac{2\mu_E}{(x_e - D_E)^3}\Delta x + \frac{2\mu_M}{(x_e + D_M)^3}\Delta x + \frac{F_x}{m}$$

$$2\left(\frac{\mu_E}{(x_e - D_E)^3} + \frac{\mu_M}{(x_e + D_M)^3} \right) \Delta x + \frac{F_x}{m}$$